

UTSAV CHAUDHARY

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Senior **DevOps Engineer** with **6 years** architecting and deploying **production grade AWS and Azure infrastructure** for high availability systems. Demonstrated success in reducing cloud expenditure and deep proficiency in **Kubernetes + Helm orchestration**, **Infrastructure as Code** with **Terraform** and **Ansible**, and **cross cloud deployments** across Dev, Test, UAT, QA, and Production. Extensive experience architecting and testing comprehensive **disaster recovery** strategies that meet stringent **RTO** and **RPO** requirements for business critical systems.

Technologies

Cloud & Infrastructure: AWS (EC2, EKS, ECS, Fargate, Lambda, RDS, S3, EBS, VPC, IAM, CloudWatch, API Gateway, Secrets Manager, Route 53, ALB/ELB, CodeDeploy, KMS, CloudHSM), Azure (App Services, AKS, Azure DevOps, Azure SQL), Docker, Kubernetes, Helm, Rancher, Argo CD

IaC & Automation: Terraform, Ansible, Puppet, Chef, Bash, Python, PowerShell, YAML, CloudFormation, Nginx, Cron

CI/CD & Version Control: Jenkins, GitHub Actions, Bitbucket Pipelines, AWS CodePipeline, Azure DevOps, Git, GitHub, GitLab, Bitbucket, AWS CodeArtifact

Monitoring & Logging: Prometheus, Grafana, Elasticsearch, Logstash, Kibana, CloudWatch, Application Insights, Datadog, Elastic Stack

Networking & Security: VPCs, Security Groups, IAM Policies, DNS, VPN, LDAP, RBAC, SSL/TLS, OAuth 2.0

Development & Tools: Python, TypeScript, JavaScript, Groovy, FastAPI, Node.js, React.js, Next.js, REST APIs, gRPC, WebSockets, PostgreSQL, MongoDB, Redis, Firebase, Kafka, Jira, Confluence

Operating Systems: Linux (Ubuntu, CentOS, RHEL), Windows Server, macOS, Raspberry Pi

Experience

Sr. DevOps Engineer, Hybrid, Labs196, TX

Feb 2025 - Present

- Architected and implemented **multicloud strategy** serving **45+ microservices** across **AWS** and **Azure** infrastructure, reducing single cloud dependency and ensuring **high availability Kubernetes clusters** with **Helm** deployment strategies, ensuring consistent environment parity across **Dev, Test, UAT, QA, and Production**.
- Operated **200+ node Kubernetes** clusters and automated application delivery with **Argo CD GitOps**, eliminating manual deployment errors and enabling synchronized releases across all valid **GxP** environments.
- Implemented **Rancher** as the centralized **Kubernetes** management platform to govern **EKS** and **AKS** clusters, and integrated Rancher with **Azure Active Directory (Entra ID)** for RBAC enforcing **least privilege** permissions and maintaining **audit-ready logs** for **FDA** and internal compliance reviews.
- Established infrastructure reliability through **Terraform** and **Ansible** governance, managing **VPCs**, **security groups**, **IAM policies**, and **RDS** lifecycle across multiple environment infrastructure with version controlled **IaC**.
- Led **observability strategy** implementing **Prometheus** and **Grafana** for infrastructure metrics, **CloudWatch Logs** and **ELK Stack** for centralized logging, and alerting, enabling rapid incident response and maintaining **99.9% SLA** compliance.
- Automated backup procedures for cloud resources including **S3** buckets, **Azure Blob Storage** and **databases** across both platforms, ensuring data durability and compliance with **GxP** retention requirements.
- Designed and integrated **edge compute** and **CDN** server infrastructure for edge deployments, reducing latency and improving content delivery aligned with project architecture.
- Participated in **on-call** rotations and **incident response**, creating **runbooks** and collaborating across teams to resolve infrastructure issues while maintaining **production** reliability.

Sr. DevOps Engineer, OnSite, Finix196, TX

Jan 2024 - Feb 2025

- Architected and deployed production **AWS EKS** clusters for **containerized workloads** behind **Application Load Balancers (ALB)**, sustaining **99.9% availability** for **trading** and **risk management** services.
- Implemented centralized **monitoring** and **alerting** with **Prometheus** and **Grafana**, integrated with **Amazon Route 53** health checks and **load balancer** metrics, reducing **mean time to detect (MTTD)** for production incidents from **30 minutes** to **5 minutes** across **real-time payment** and **settlement** systems.
- Led migration of legacy applications from **AWS EC2** to **ECS Fargate**, eliminating server management overhead and improving **resource utilization** by **40%** for the **customer onboarding** and **KYC** platform.
- Deployed **canary release** strategy using **AWS CodeDeploy** with **Application Load Balancer (ALB)** traffic shifting, reducing deployment related incidents by **75%** for the **online banking portal**.
- Automated **backup** and **disaster recovery (DR)** for **Amazon RDS** databases and **EBS** volumes, meeting **RPO** of **10 minutes** and **RTO** of **30 minutes** for critical **transaction processing** systems.

- Architected secure **data encryption** and **key rotation** using **AWS KMS** and **AWS CloudHSM**, protecting sensitive **financial data at rest** and **in transit** while achieving compliance with internal **cryptographic standards**.

DevOps Engineer CO-OP, Midh Technologies, TX

June 2023 - Dec 2023

- Managed production infrastructure for ML workloads, supporting **GPU backed Kubernetes clusters** on **AWS EKS** with cost optimization strategies including node pool configuration, resource quotas, and auto scaling policies.
- Designed high-performance inference infrastructure deploying **FastAPI** endpoints behind **AWS API Gateway** with **IAM authentication**, request throttling, and **Lambda** integrations for scalable ML inference.
- Developed infrastructure automation through **Ansible playbook** development, reducing manual server configuration and deployment time by **50%** while ensuring configuration consistency.
- Built production **CI/CD pipelines** with **GitHub Actions**, automating testing, Docker image builds, security scanning, and deployments to **AWS Amplify** and **EKS** environments.
- Implemented infrastructure security practices including **RBAC** policies, **IAM role** management, and **VPC** security configurations across AWS services and application layers.

DevOps/Platform Engineer, Compendious Medialabs Pvt. Ltd., IN

June 2021 - July 2022

- Managed hybrid infrastructure for **on premise Linux servers** (Ubuntu) with containerized deployments using **Docker**, establishing infrastructure reliability and automation through **Bash** scripting and configuration management.
- Designed and implemented self hosted CI/CD infrastructure deploying **GitHub Actions runners** on local infrastructure, building production grade pipelines for automated testing, Docker builds, and multiple environment deployments.
- Established production grade reverse proxy infrastructure with **Nginx** and **SSL/TLS** termination, optimizing external service exposure with **250ms** average response time and high availability.
- Implemented observability strategy deploying **Prometheus** and **Grafana** monitoring on local infrastructure; automated infrastructure tasks with **Python/Bash** scripting, reducing manual operational overhead by **60%**.

Software Engineer / DevOps Remote, VRoKCs, MO

March 2020 - Sept 2020

- Managed production infrastructure on **Azure**, deploying **Flask web applications** on **Azure App Services** with high availability configurations, auto scaling, and integrated monitoring.
- Built infrastructure automation through **Azure DevOps** pipelines for ML model deployment, implementing artifact storage with **Azure Blob Storage** and version control workflows.
- Designed containerized infrastructure using **Azure Container Instances** and **Docker** for ML inference services, ensuring environment parity between development and production with consistent deployment practices.
- Developed CI/CD infrastructure using **Azure DevOps** for ML lifecycle management, automating training and deployment workflows and reducing deployment cycles by **40%** through pipeline optimization.

Web/Game Developer, underDogs Gaming Studios, IN

March 2019 - Aug 2019

- Contributed to backend API infrastructure for gaming services, integrating and optimizing **Quarters API** across multiple games with secure transaction, supporting **load balancing** and **API rate limiting** for high availability systems.
- Supported infrastructure reliability initiatives by optimizing real time client server communication, validating and refining backend interaction flows using **Python** and **C#**, implementing error handling and health monitoring.
- Developed unified transaction infrastructure connecting external services (**payments, inventory, shipping**) to internal game systems, following consistent API patterns and infrastructure best practices.
- Participated in infrastructure security and testing practices, implementing **API security best practices**, comprehensive logging, and integration testing frameworks for edge cases including payment failures and network resilience.

Education

State University of New York at Binghamton

Master of Science in Computer Science

Aug 2022 - Dec 2023

3.44/4.00 GPA

Certifications



Microsoft Certified: DevOps Engineer Expert

AZ-400: Designing and Implementing Microsoft DevOps Solutions

AZ-900: Microsoft Certified: Azure Fundamentals

Google IT Automation with Python - (Specialization)

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